

RYAN M. KIM

kim.ryan8@northeastern.edu | [linkedin.com/in/ryan-kim-865356324](https://www.linkedin.com/in/ryan-kim-865356324) | ryanmkim.github.io/ryankim

EDUCATION

Northeastern University

Expected 2028

B.S. in Electrical and Computer Engineering, Dean's List

Boston, MA

- **Coursework:** Digital Design and Computer Organization, Embedded Design: Enabling Robotics, Discrete Structures, Circuits and Signals, Fundamentals of Algorithms, Fundamentals of Networks, Physics I/II, Calculus III, Differential Equations and Linear Algebra

TECHNICAL SKILLS

Programming: Python, C/C++, Racket, Bash, MATLAB, PyTorch, scikit-learn, NumPy, Pandas

Hardware: Vivado, Quartus Prime, SystemVerilog, Verilog, LTspice

Tools: Git, Linux, LaTeX, Markdown, Oscilloscope, Multimeter

RESEARCH EXPERIENCE

Northeastern University Computer Architecture Research

Jan. 2026 – Present

Undergraduate Researcher

Boston, MA

- Implemented optimized matrix multiplication using tiling techniques to improve cache locality and computational throughput
- Developed multithreaded matrix multiplication programs using POSIX Threads (pthreads) in both C and C++
- Applied synchronization primitives including mutexes to manage shared memory access and ensure thread-safe execution in concurrent programs

Northeastern University – Prof. Yanzhi Wang

Dec. 2025 – Present

Undergraduate Researcher

Boston, MA

- Researching world models for robotic manipulation tasks to improve model-based planning and decision-making in complex environments

UMass Chan Medical School, Dept. of Biochemistry and Molecular Biotechnology

Aug. 2022 – Sep. 2024

Research Assistant, Schiffer Lab

Worcester, MA

- Investigated structural basis of influenza virus entry using cryo-electron tomography and deep learning
- Trained convolutional neural networks to automatically segment viral protein components, processing 100+ tomograms and enabling subnanometer structural determination of 12,000+ particles at 8–10 Å resolution
- Contributed to quantitative morphology analysis pipeline using scikit-learn, Open3D, and EMAN2.99

PUBLICATION

Q.J. Huang, R. Kim, K. Song, N. Grigorieff, J.B. Munro, C.A. Schiffer, & M. Somasundaran. "Virion-associated influenza hemagglutinin clusters upon sialic acid binding visualized by cryoelectron tomography." *Proc. Natl. Acad. Sci. U.S.A.* 122(16): e2426427122 (2025).
doi.org/10.1073/pnas.2426427122

PROJECTS

Sonar FPGA Ranging System | *Quartus, FPGA*

- Implemented a custom hardware driver in SystemVerilog using Quartus to interface an HC-SR04 ultrasonic sensor with a DE1 FPGA board.
- Designed synchronous sequential logic counters driven by a 50 MHz clock to generate trigger pulses and capture returning echo durations for precise distance measurement.
- Developed hardware logic to calculate acoustic round-trip travel times and convert clock periods into centimeters, outputting real-time data to 7-segment displays.
- Verified system timing and debugged module performance by utilizing an ADALM 2000 oscilloscope to analyze input/output pulse widths against physical test targets.

Parallel Tiling Image Filters | *C++, POSIX Threads*

2026

- Built a multithreaded image processing tool that splits images into tiles and applies convolution-based filters using a thread pool architecture
- Implemented grayscaling, thresholding, Gaussian blur, edge detection, and sharpening
- Implemented thread-safe task scheduling with POSIX mutexes and condition variables

WM-811K Wafer Defect Analysis Pipeline | *Python, Pandas, NumPy, SciPy*

2025

- Developed a semiconductor defect analysis pipeline in Python to process 172,000 wafer maps across 8 classes, utilizing SciPy for image interpolation and resizing to generate averaged probability heatmaps
- Engineered an automated reporting system using Matplotlib and fpdf2 to visualize statistical trends and generate a 20-page defect "field guide"

ENGAGEMENT EXPERIENCE

Youth For Community Improvement

2021 – 2023

Grant Committee Member

Worcester, MA

- Evaluated grant applications and allocated \$40,000 in funding to nonprofit organizations addressing community issues through a youth-led program, facilitating meetings with community speakers and authoring advocacy briefs to inform grant decision-making